

AUTOMATIC DOOR CONTROL WITH PLC PROGRAM

This project designs the Automatic Sliding Door Control with PLC to automatically open and close a sliding door using sensors, a motor, and a PLC controller, improving convenience, safety, reliability, and efficient door operation.

Aim

- Design and implement an automatic sliding door using a programmable logic controller (PLC).

Objectives

- Study the fundamental components of an automatic sliding door.
- Enable automatic opening and closing via sensors, eliminating the need for manual operation.
- Improve energy conservation and security in buildings through the practical application of modern technology.

Methodology

System Design → Component Selection → Hardware Installation → Sensor and Motor Connection → PLC Programming → System Integration → Testing and Troubleshooting → System Performance Evaluation



HYDRO POWER GENERATION SYSTEM

This project designs a prototype validation of an 85%-efficient Pitchback waterwheel micro-hydro system with 5 V DC generator for off-grid residential electricity.

Aim

- To provide a sustainable, renewable, and locally sourced energy solution that meets the electricity needs of households.

Objectives

- Site assessment and design of a 1 W Pitchback waterwheel for household electricity.
- Pair with a 5 V DC generator and running simulations to model.
- Build a functional prototype to measure efficiency and promote renewable water energy.

Methodology

Site assessment and data collection → Pitchback waterwheel design → 5 V DC motor integration → Computational simulation → Prototype fabrication and performance testing.



AUTOMATIC FLOOR CLEANING MACHINE

This project is designed to automatically clean floors, uses an Arduino system to control the motors and sensors, enabling the cleaning machine to detect and avoid obstacles while cleaning the floor efficiently.

Aim

- To design and develop an automatic floor cleaning machine that can efficiently remove dust and particles from the floor with minimal human effort.

Objectives

- To develop an Arduino-based automatic floor cleaning machine.
- To detect and avoid obstacles using an ultrasonic sensor.
- To reduce human effort, cleaning time, and physical workload through automation.

Methodology

Project Design → Arduino Programming → Sensor and Motor Wiring → Automatic Cleaning and Obstacle Avoidance → Testing and Power Supply → Performance Evaluation



AUTOMATIC SOLAR PANELS DUST CLEANING AND TRACKING USING PLC

This project presents the design and implementation of an automatic single-axis solar tracking and cleaning system using PLC.

Aim

- Construct single-axis solar tracking and cleaning system using PLC.

Objectives

- Understand the operation of a single-axis solar tracking and automatic cleaning system.
- Study how the system adjusts the solar panel position and remove dust to improve energy efficiency.
- Understand the PLC-based control process, including sensor inputs, program logic, and automatic actuator operation.

Methodology

System requirement analysis → PLC programming → Mechanical structure assembly → System integration and testing → Performance evaluation.



DESIGN AND CONSTRUCTION OF INTELLIGENT MOBILE TRAFFIC LIGHT SYSTEM USING PIC MICROCONTROLLER

This project presents the design and development of a portable, intelligent traffic light system controlled by a PIC microcontroller to manage temporary traffic control, and dynamically reduce roadway congestion.

Aim

- Design and construct a mobile intelligent traffic light system utilizing a PIC microcontroller.

Objectives

- Implement real-time traffic signal timing adjustments based on traffic density inputs.
- Provide a portable, self-powered mobile solution for temporary roadworks, emergency zones, or event traffic control.
- Enhance traffic flow management and safety while minimizing manual traffic officer interventions.

Methodology

System requirement design → PIC hardware and circuit interfacing → Basic pro language programming → Prototype assembly → Field testing & evaluation.

